

HUGHENDEN, Australia

WMO: 943430

Lat: 20.819S

Lon: 144.233E

Elev: 317

StdP: 97.57

Time Zone: 10.00 (E10)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.9	6.7	-10.7	1.6	17.3	-9.3	1.8	17.6	9.0	19.7	8.3	20.1	1.3	150	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	12.4	39.4	19.3	38.2	19.2	37.1	19.2	24.2	30.0	23.7	29.4	23.2	29.0	4.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.1	26.3	22.2	17.5	26.3	21.6	16.9	26.3	75.0	30.0	72.8	29.3	70.8	28.6	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	8.2	7.5	DB	1.9	41.0	1.3	1.5	1.0	42.1	0.2	43.0	-0.5	43.8	-1.5	44.9	(4)
(5)			WB	-1.3	25.4	1.7	1.0	-2.6	26.1	-3.5	26.7	-4.5	27.3	-5.7	28.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.7	29.2	28.5	27.7	25.2	21.4	18.1	17.9	19.3	23.6	26.8	28.8	30.0	(6)	
	DBStd	5.15	2.20	2.18	1.89	2.21	3.18	3.39	3.46	3.44	3.05	2.92	2.25	2.20	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	143	0	0	0	1	12	46	52	31	2	1	0	0	(9)	
	CDD10.0	5361	596	518	549	456	353	243	244	288	407	522	563	621	(10)	
	CDD18.3	2462	338	284	291	206	106	39	38	60	160	263	313	363	(11)	
	CDH23.3	30403	4322	3498	3408	2106	918	313	358	756	1955	3502	4181	5087	(12)	
(13)	CDH26.7	15714	2341	1838	1700	903	262	47	59	228	924	1959	2420	3033	(13)	
(14)	Wind	WSAvg	4.1	4.2	4.2	4.3	4.2	3.9	3.6	3.7	4.1	4.4	4.4	4.3	(14)	
(15) Precipitation	PrecAvg	449	104	96	53	23	18	14	9	9	6	22	38	68	(15)	
	PrecMax	922	629	283	252	237	112	107	113	119	77	193	171	226	(16)	
	PrecMin	240	0	1	0	0	0	0	0	0	0	0	0	5	(17)	
	PrecStd	178	111	74	60	45	27	27	21	22	18	38	43	59	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	39.5	37.5	34.8	33.0	29.8	30.3	33.2	36.8	39.5	39.8	40.9	(19)	
		MCWB	20.4	20.2	19.9	19.5	18.3	17.4	17.1	16.6	16.5	18.0	18.9	19.5	(20)	
	2%	DB	38.3	37.2	36.0	33.5	30.8	28.0	28.2	31.0	35.0	37.6	38.3	39.6	(21)	
		MCWB	20.3	20.4	20.0	18.8	17.4	16.8	16.0	15.9	16.4	17.6	18.6	19.3	(22)	
	5%	DB	36.7	35.8	34.7	32.4	29.4	26.6	26.8	29.2	33.2	36.2	37.1	38.2	(23)	
		MCWB	20.5	20.7	20.0	18.3	16.9	16.0	15.2	15.0	16.1	17.4	18.5	19.4	(24)	
	10%	DB	35.2	34.6	33.6	31.3	28.0	25.2	25.4	27.5	31.7	34.8	35.9	36.9	(25)	
		MCWB	20.6	20.6	20.0	18.1	16.5	15.3	14.5	14.2	15.8	17.3	18.3	19.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.8	24.9	24.0	23.1	20.1	19.5	19.0	19.1	20.8	21.0	23.5	24.0	(27)	
		MCDB	31.1	31.1	29.5	28.2	28.3	26.0	25.0	28.3	27.4	29.7	29.9	30.0	(28)	
	2%	WB	24.2	24.2	23.2	21.8	19.4	18.2	17.8	17.6	19.7	20.0	22.2	23.2	(29)	
		MCDB	30.2	30.0	29.0	28.2	26.8	24.8	25.4	26.3	27.2	30.8	28.5	29.7	(30)	
	5%	WB	23.7	23.6	22.6	21.0	18.6	17.4	16.8	16.5	18.4	19.4	21.4	22.7	(31)	
		MCDB	29.5	29.3	28.8	27.6	26.0	24.2	24.0	25.7	27.5	30.4	29.0	29.6	(32)	
	10%	WB	23.2	23.1	22.0	20.2	17.8	16.6	15.8	15.5	17.2	18.8	20.7	22.1	(33)	
		MCDB	28.9	28.7	28.7	27.2	25.4	23.2	23.2	24.5	28.0	30.0	29.4	29.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	10.9	11.3	11.9	12.9	13.8	14.8	15.9	15.4	14.7	13.2	12.4	(35)	
		MCDBR	13.4	13.3	13.2	13.4	13.7	14.1	15.0	16.7	17.1	16.3	14.9	14.7	(36)	
		MCWBR	3.5	3.5	3.7	4.3	5.1	5.7	6.2	6.6	6.0	5.4	4.4	4.3	(37)	
	5% WB	MCDBR	9.4	9.4	9.7	9.8	10.4	11.4	11.8	13.0	12.4	13.0	11.5	10.9	(38)	
		MCWBR	2.7	3.0	2.9	3.3	4.0	4.5	4.8	4.8	4.3	4.0	3.6	3.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.375	0.368	0.352	0.337	0.320	0.308	0.300	0.316	0.350	0.375	0.383	0.392	(40)	
	taud		2.534	2.552	2.577	2.580	2.562	2.587	2.592	2.520	2.434	2.382	2.409	2.428	(41)	
	Ebn at Noon		969	963	955	930	908	902	922	938	942	945	955	953	(42)	
	Edh at Noon		112	108	101	94	88	82	84	98	115	127	126	124	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.92	6.86	6.38	5.74	4.81	4.31	4.79	5.72	6.72	7.33	7.50	7.32	(44)	
	RadStd		0.56	0.48	0.54	0.50	0.29	0.34	0.22	0.27	0.40	0.45	0.48	0.53	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page